

Spark Performance Optimization

Bucketing

Partitioning isn't always enough.

Bucketing helps Spark perform faster joins by organizing data into fixed buckets.



What is Bucketing?

Bucketing divides data into a fixed number of files using a hash function.

Unlike partitioning, bucket values don't create directories. Spark stores records with the same bucket value together.

- ✓ Faster Joins
- ✓ Less Shuffle
- ✓ Better Query Performance

Why is it Important?

Bucketing helps:

- ✓ Reduce Shuffle During Joins
- ✓ Faster Aggregations
- ✓ Better Data Distribution
- ✓ Improved Query Performance
- ✓ Efficient Large Table Joins

When Should You Use Bucketing?

Bucketing is useful when:

- ✓ Frequently Joining Large Tables
- ✓ Repeated Analytical Queries
- ✓ Large Fact Tables
- ✓ Data Warehouse Workloads
- ✓ Static Datasets

Common Mistakes

Bucketing may not improve performance because of:

- ✓ Different Number of Buckets
- ✓ Different Bucket Columns
- ✓ Small Datasets
- ✓ Frequent Data Updates
- ✓ Poor Bucket Selection

Choose bucket columns with high cardinality.

Best Practices

For better performance:

- ✓ Bucket Large Tables
- ✓ Use Same Bucket Count
- ✓ Bucket on Join Columns
- ✓ Combine with Sorting
- ✓ Monitor Query Plans

Real Production Example

A reporting pipeline joined customer and transaction tables multiple times every day.

After bucketing both tables on **Customer_ID**:

- ✓ Shuffle reduced significantly
- ✓ Join execution became faster
- ✓ CPU utilization improved
- ✓ Daily reports finished much earlier

Key Takeaway

Bucketing organizes data for faster joins.

The goal is simple:

- Less Shuffle
- Faster Queries
- Better Data Distribution
- Improved Join Performance
- Faster Analytics

Save this carousel for future reference.

Have you used Bucketing?

- Yes
- No
- Planning To
- Learning Spark